ICT Unit E: Social Implications

Ch25 Equity, Work and Health Issues

➢ Equity of access

➥ **Information age**

- Characterized by the fact: We can **easily** obtain a **large** amount of information **from the internet**

Advantages	Disadvantages
1. **Various** kind of information is provided	1. **Uncensored** 未經審查 information is too much
2. **Free of charge** information is the majority	**2. Unreliability, Fake, Dangerous**
3. **Up-to-date** information is available	3. **Addictive** to the internet

➥ **Digital divide**

- **Growing gap** between those people who can (vs cannot) Access + Made effective use of ICT
- **Global digital divide**: Difference in **technology level** among countries
- **Reasons**:
  1. Socio-economics (e.g. wealth)
  2. Race (e.g. black people)
  3. Geographical areas (e.g. rural and urban areas)

HKSAR	Charity group
1. ($) Provide free internet access **in public places**	1. ($) **Organize compaigns** → Collect + Mend(repair) old computers
2. Organize ICT training courses **for non-internet users**	2. ($) **Donate** to those who can't afford new ones

➥ **Globalization**

- Rapid ↑ in **International trade** + **Cultural exchanges** → Changes in society and the world economy
- **Advancement of IT** → Pace of globalization

➥ **Knowledge-based society** (Facilitated by the IT)

- In which people need to **constantly improve** themselves through **lifelong learning**
- Enrich people Culturally + Materially → Sustain the society

➥ **Gender inequity**

- All people **regardless their gender** should **Equally prepared** + **Equally engaged** in the use of technology
- To reduce gender inequity
  1. Attract more women into **Computer science + Related technical fields**
  2. Make female **more comfortable** in the **Culture of technology**
  3. **Better UI(user-interface) design** → Cater both gender

➥ **Ease of access** (Examples)

- Software: The Grid 2 (Voice control + Voice → PC Control) JAWS (Text → Voice), Dragon Naturally Speaking (Voice → Text)
- Hardware: LOMAK Keyboard (Using Laser to control PC)

Carousell@dse_note_2025xx

➢ New types of offices

1. **Telecommuting**
   - **A work arrangement**: Employees work away from a company's standard workplace
   - **Work**: Report [Progress of projects + Assignments + Problems] experienced to supervisors

	Advantages	Disadvantages
To employees	1. ↓ $ + Time 2. Δ Working time > 0	1. Long working hour 2. Work alone 3. Cannot meet colleagues 4. Equipment (self-sufficiency)
To employers	1. ↑ Productivity 2. ↓ $	1. Security risk 2. Discipline problems

2. **SOHO** (Small Office & Home Office)
   - Category of companies: Consist of 1~ 10 employees
3. **Virtual organization**: Bridge people from anywhere to work together
   - Conduct meeting through video conferencing (Tele-centre, Satellite office)

➢ Work and Health issues

- Health issues
  1. RSI (**Repetitive strain injures**)
     - Prolonged repetition of physical movement (e.g. wrist pain)
  2. CVS (**Computer vision syndrome**)
     - Prolonged exposure to EMR (Electromagnetic radiation) (e.g.) Glaucoma 青光眼
- Preventions
  1. **Ergonomics**
     - Science of integrating **Efficiency + Safety + Comfort** into workplace design
       ⇒ Study of **physical relationship** between human and their tools
     - Ergonomic Devices:
       i. **Image-talk symbol writer** (For Speech disordered people)
       ii. pVoke Software (For those who cannot speak)
  2. **Health guidelines for computer users**
     - Posture & Duration of using computer
     - Lighting environment when using the computer

     Working area + Alternate work activities + **Breaks** + **Stretching exercises** + ↓ EMR+ ↓ Surrounding noise + RSI + CVS + Internet addiction
     → Social problem
     → Impairment of real life relationships and financial problems

Carousel@dse_note_2025xx

Ch26 Intellectual Property

➢ Intellectual property rights (IPRs)

- The rights granted to investors, writers, … and other talents to **protect their inventions & products**
- IPRs consist a group of intangible 無形 property right

Copyrights	Patents 專利	Design	Trademark
Original work	Inventions	Appearance + Design	Logo, Sign

- Copyrights
  - **Automatically granted** to the owner after creation
  - **Copyright infringement** (Plagiarism): Distribute/ Download copyright material on the internet **without Owner's permission**
  - **Copyright Ordinance** 版權條例 (97'): Prohibits unauthorized **Duplication + Use** of computer programmes

➢ Licences and types of software

- Licences of software (Legal contract: Define the way the user can use the program)

➥ **Single-user licence**

- For 1 single and separated unit

➥ **Multi-user licence** (Site licence) e.g. Network licensed software (Installed on server) (Cloud)

- For 1 corporation (Cost depends on number of employees, ✕ the locations) → 1 unit → 1 site licence is needed

Types of software

Source code disclosure	Types of software	Features
Open source	Public domain software	• $0: Donated to the public use • No copyrights restriction
Either	Freeware	• $0 + Unlimited use • May allow free distribution
Closed source (Proprietary software) (專有軟件)	Shareware	• $0 within trial period • May lock functions
	Commercial software	• $ > 0 • Better maintenance + support

➢ Piracy & measures to strike

➥ **Definition and meaning** (piracy = 盜版)

- Software piracy: **Unauthorized** + **Illegal** duplication of copyright software
- Internet piracy: **Transfer** (Distribute/ Download) unauthorised digital through internet
- Illegal copies: **Pirated software** (Through File sharing web sites/ Peer-to-peer file sharing services)

↑ Use of pirated software → ↓ Profit + Diminish software industry

Carousel@dse_note_2025xx

➥ **Measures to strike** (↓ Property theft 盜竊)

Strength	Measures	Features
 Best	**Digital watermark**	Technique: Add hidden copyright notices to digital files
	Licence agreement	Statement: States the right and restrictions users have to follow
	Online CD key	Software activation process: Prevent multiple copies of the same key
	Machine ID	Prevent multiple installations on different computers (e.g. Windows XP) → Machine Dependent
	Hardware key (dongle)	Hardware device: Connect and authenticate the software

- **Digital signature** (for transmission of data)
  - There are 2 ways to encrypt the data before transmission:
    1. Tom encrypts using Amy's **Public key** → Amy decrypts using Amy's **Private key**
       To ensure that Amy is the sole receiver (˙.˙Decrypt with Amy's private key)
    2. Tom encrypts using Tom **Private key** → Amy decrypts using Tom's **Public key**
       To ensure that Tom is the sole receiver (˙.˙Encrypt with Tom's private key)

Tom (sender) → Transmission → Amy (receiver)

*Using Encryption / Decryption key of the same user

Remainder: Tom cannot have Amy's private key ; Amy cannot have Tom's private key

Ch27 Threats and Security on the internet (I)

➢ Security threats on the internet

➥ **Malware** (Malicious software)

- Causes **damage or denial of services (DoS)** on infected computer system
- **Spread through email, storage devices or internet**
- Hidden method: Divide and encrypt back to a program

1. **Computer virus** (Part of program)
   - **Attached** itself to other normal programs
   - **Replicate** itself after its host program runs → Infect other computer system
   - **Macro virus**: Virus written in MS Office Macro language (Hide inside a document)
2. **Virus hoaxes** (Email) (e.g. Spam)
   - **Warning** of non-existent virus
   - Sent **through email**
   - **Urge** 促使 user to forward the letter
3. **Worm** (Computer program)
   - Uses **Computer networks** + **Security loopholes** to spread
   - **Replicate** itself automatically
   - **Method**: Scan the address book → Send through email
4. Trojan horse (Computer program)
   - Perform **malicious/ destructive** actions
   - **Disguises** 偽裝 itself as a useful program
   - **Doesn't replicate**

Carousel@dse_note_2025xx

- Used by **hacker** to **steal sensitive information**
- May allow hacker to **control computers remotely**

➥ **Spyware**

• Hides itself + Look for personal data (e.g. Google_Redirect_Virus)

➥ **Adware** (Advertisement in Live Messenger)

• Keeps track of user's activities → Display related advertisement

➢ Security application software

1. **Anti-virus software**
   • **Jobs**: Examine files **stored on a disk/ downloaded files** → Determine if they are infected + disinfect if needed
   • **Functions**: Monitor **running program** + Stop **suspicious instructions** + Terminate running programs
   • **Virus signature** (Virus definitions): Identify known virus
   → **Scanning files**: Search for virus definitions + Report infected files to the user
2. **Anti-spyware program**
   • **Jobs**: Examines **downloaded webpages** + **Executable files**
   • **Functions**: Detects system **abnormalities** (e.g. Arbitrarily opened network channel)
3. **Firewall software (Computer application)**
   • **Jobs**: Examines the **network traffic** → Authenticate transmission
   • **Functions**: Safeguard **all incoming + out-going data** (Through the internet)
4. **Hardware firewall**
   • **Jobs**: Provide more **Comprehensive + Reliable network monitoring services**
   • Located between LAN and WAN

➢ Browsers and security

1. **Plug-in** (ActiveX control, Java Applets)
   • Browser **denies**:
   - **Execution** of these **unidentified components**
   - **Alert** users when web sites **request downloading**
   • Browser **identifies**:
   → ID of component using **digital signature** (e.g. Public key of the software developer)
2. Malicious scripts (by plug-in)
   • **Exhaust** 耗盡 system resources + Eventually **bring down** the whole system
   • Hackers may **exploit the vulnerabilities** of web browsers → Control computer for illegal purposes
3. AJMAX (Asynchronous JavaScript And XML) (Example of Plug-in)
   - Make **server-side queries**
   - On-the-fly (**Remote**) display
4. Firewall
   - Analyse IP address + Types of traffic
   - Judge permissions of I/O of dat
   - Address translation (IP)

Carousel@dse_note_2025xx

Ch28 Threats and Security on the internet (I)

➢ Data privacy

- Public expectation of privacy in the **Collection + Holding + Processing + Use of personal data**
- **Right of privacy**: Fundamental human right
- When providing personal data, we have rights to …
    - **Know** the **kind + purpose** of the collected data
    - Access (Know **the medium of storage**) + **Amend** the collected data

➢ Privacy threats on the internet

- **Hackling**: **Illegal access to data** in a computer system / computer network
- **Hackers**: Skillful individuals on **Computer security** + **System exploits**
  → **Break security controls** of a computer network → **Gain unauthorized access**
  → Using **Sniffers**: **Capture** + **show the data transmitted** through the internet
  (Originally used to monitor network traffic)
  → ✕ Steal sensitive information
- **Crackers**: **Criminal hackers** →**Steal sensitive information** → Hack for illegal activities (Have **commercial interest**)
- **Spam** (Junk-email): **Unsolicited** 不請自來 email / **Advertisement** of commercial products
    - Email filtering: e.g. MailWasher
- **Phishing**: Sending email message (Seems to be sent from trustworthy entities / organizations / banks)
    - **Similar URL** → Attempts to **fake and get confidential data** (e.g. credit card numbers)

➢ Ways to maintain privacy

- **Being anonymous**
    - To prevent **identification of user's location** (IP address) and **personal information** (ISP)
    - **Proxy server** (Computer ⇔ Server): Route data packets through > 1 computers
- **Passwords**
    - Strong (> 8 characters)
    - ✕ Copies / Share
    - Change it regularly + Unique
    - Aware of fake email / websites and suspicious call
    - Security software
- **Internet user's protection**
    - Scan email / files
    - ✕ Cookies
    - ✕ File sharing
    - ✕ doubtful program installation
    - ✓ Patches 補丁

Carousel@dse_note_2025xx

➢ Encryption technologies

- **Encryption**: Conversion with algorithm: Readable data (Plaintext) → Unreadable character (**Cipher-text**)
- Length of encryption key: **Difference effectiveness** of encryption + **security level** (e.g. 128bit is common)

1. Symmetric key encryption
   - Employs the same key → Encode + Decode the message
2. Public key and private key cryptography (Digital signature)

➢ Access control – authentication and authorization

1. **Authorization** (Level of permission)授權
   - Use ACLs (Access Control List): Precisely record the security permission (Log-on information + Mode of access to files)
2. **Accounting** (Log keeping + Activity recording)
3. **Authentication** (Identification of someone + compare database record)認證
   - **Username-password approach**: Common way
     (**Security string** – low level access control: ✕ Username ✓ Password for specific group)
   - **IP address**: but **unstable** (˙.˙ changeable) + **shareable**
   - **Digital Certificate** (Computer file): Contain **electronic signature** → **Confirm id of an individual** (Public/Private key) → Accept as **personal signature**

➢ Securing transactions on the internet

1. **Secure Socket Layer** (SSL) (OSI model: Application Layer)
   ⇒ Establish a secure and encrypted communication channel: **Users ⇔ Web servers**
   ⇒ **HTTPS** (Encrypted with SSL): **Secured with digital certificate** (✕ HTTP: Plaintext only)
   ⇒ But: **Slow** encryption + decryption → **Time delay** (˙.˙ Not suitable for online gamming, chat)
2. **Security tokens** (Small hardware device)
   - Generate **numeric string** (Special mathematical algorithm)
   - Time-synchronized **one-time password**
3. **Mobile short message service** (SMS)
   - Use for **once**
   - **Expires within a short period of time**
4. Smart cards
   - **Embedded** with an **electronic chip**

Carousel@ose_note_2025xx